

STAR BRIGHTNESS AND DISTANCE

STARS DIFFER HUGELY IN THEIR BRIGHTNESS AND IN THEIR DISTANCE FROM EARTH, ALTHOUGH ALL, APART FROM THE SUN, ARE EXTREMELY REMOTE. HOW BRIGHT A STAR LOOKS FROM EARTH DEPENDS OF COURSE PARTLY ON HOW FAR AWAY IT IS.

Because stars are so far away, obtaining data about them is tricky. Most of the data about any star comes from studying the light and other radiation coming from it, while the distance to the least remote stars can be worked out by measuring tiny annual variations in their sky positions.

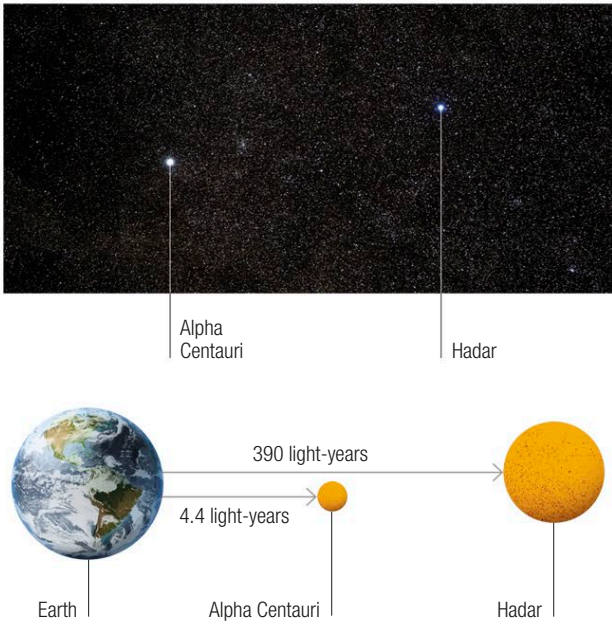
Brightness

There are two different ways of stating a star's brightness: apparent magnitude, which indicates how bright a star looks from Earth, and absolute magnitude, which expresses how bright it would look from a set distance—a better indicator of how brilliant it truly is. On both scales, a change of +1 on the scale means a decrease, and a change of -1 means an increase, in brightness. So, on the apparent magnitude scale, stars just visible to the naked eye score +6 or +5, while very bright stars score about +1 to 0, and the four very brightest have negative scores. The absolute magnitude scale runs from around +20 for some exceptionally dim red dwarfs to around -8 for the brightest supergiant stars. A star's absolute magnitude is related to a measurement called its visual luminosity. This is the amount of light energy that a star emits per unit of time. Luminosity is often stated relative to that of the Sun.

▽ Brightness comparisons

The apparent and absolute magnitudes, and luminosities, of 11 different stars, including the Sun, are compared in the table below. The stars range from the relatively nearby red dwarf, Proxima Centauri, to distant but fantastically luminous supergiants, such as Rigel.

MAGNITUDE AND LUMINOSITY OF SELECTED STARS				
Star (Constellation)	Distance from Earth	Apparent magnitude	Absolute magnitude	Visual luminosity (number of Suns)
The Sun	92,960,000 miles (149,600,000 km)	-26.74	4.83	1
Sirius A (Canis Major)	8.6 light-years	-1.47	1.42	23
Alpha Centauri A (Centaurus)	4.4 light-years	0.01	4.38	1.5
Vega (Lyra)	25 light-years	0.03	0.58	50
Rigel (Orion)	780–940 light-years	0.13	-7.92	125,000
Hadar (Centaurus)	370–410 light-years	0.61	-4.53	5,500
Antares (Scorpius)	550–620 light-years	0.96	-5.28	11,000
Polaris (Ursa Minor)	325–425 light-years	1.98	-3.6	2,400
Megrez (Ursa Major)	58 light-years	3.3	1.33	25
Mu Cephei (Cepheus)	1,200–9,000 light-years	4.08	-7.63	96,000
Proxima Centauri (Centaurus)	4.2 light-years	11.05	15.6	0.00005



△ **Apparent magnitude**
The two brightest stars in the photograph at the top—Alpha Centauri (left) and Hadar (right)—appear roughly as bright as each other. In other words, they have a similar apparent magnitude. But intrinsically, Hadar is much brighter because its absolute magnitude is greater. Alpha Centauri looks about as bright as Hadar only because it is about 90 times closer.



△ **Proxima Centauri**
This photograph is of a red dwarf star, Proxima Centauri. At 4.2 light-years away, it is the closest star to Earth other than our Sun. Although brilliant in this Hubble Space Telescope image, relatively speaking it is a dim star with an absolute magnitude of +15.6 and a luminosity that is only a tiny fraction of that of the Sun.